Exp – 1

```r
num1 <- as.integer(readline(prompt = "enter the first number:"))
num2 <- as.integer(readline(prompt = "enter the second number:"))
num3 <- num1 + num2
print(num3)

```

5:1 (Top Level) R Script

Console Terminal Background Jobs

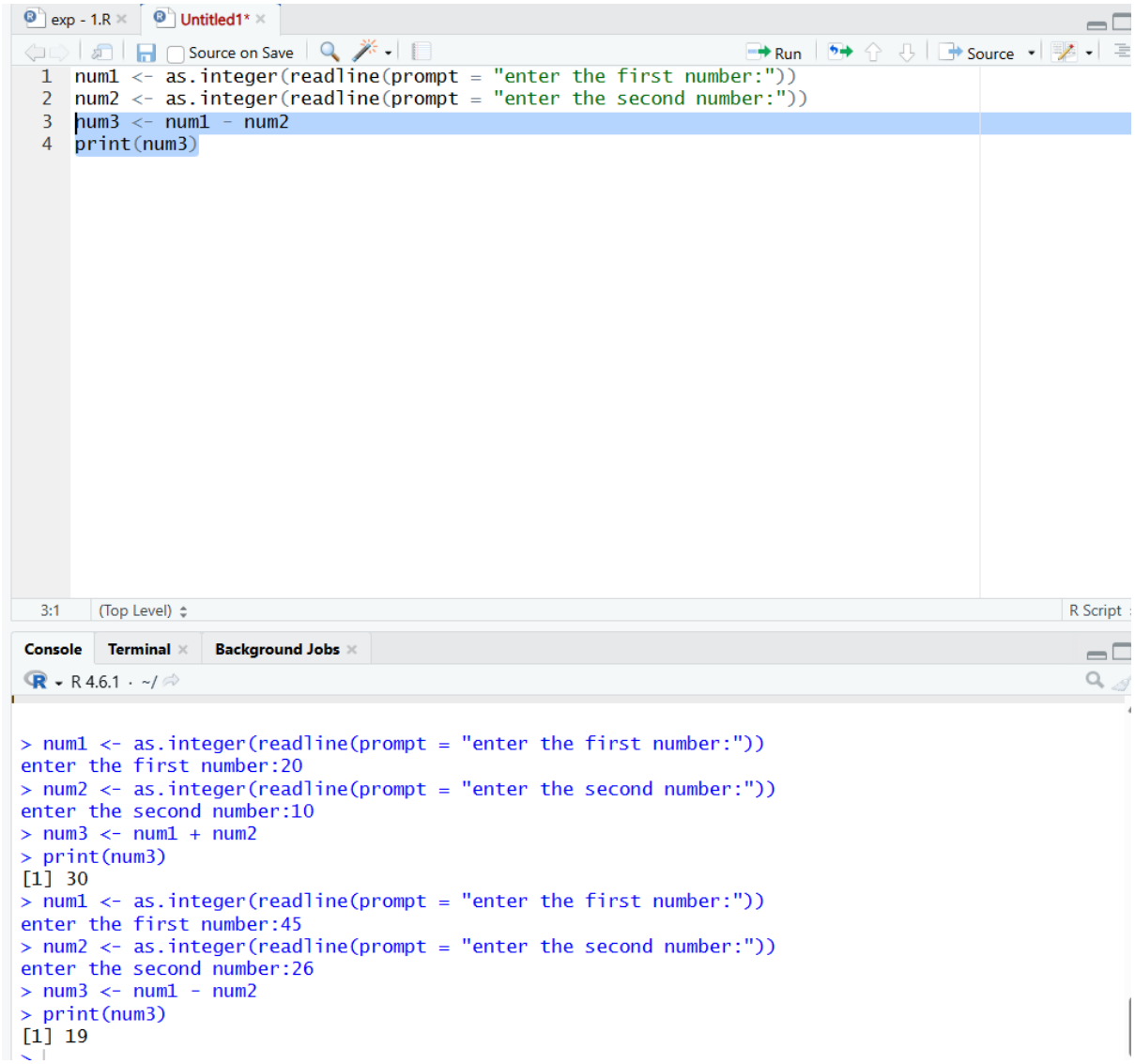
R 4.6.1 · ~/

```
> print(num3)
[1] NA
> num1=as.integer(readline(prompt = "enter the first number:"))
enter the first number:print(num3)

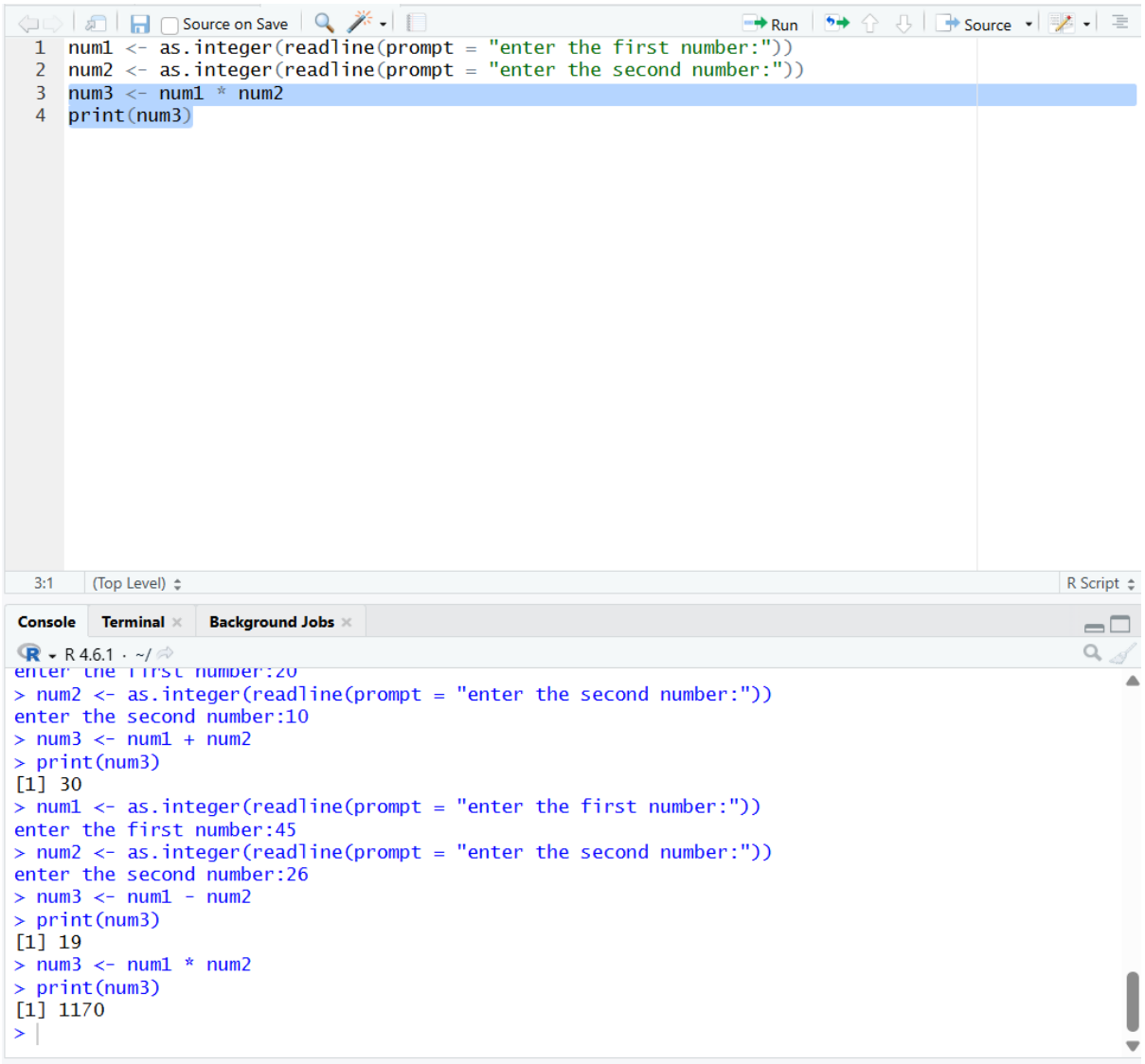
Warning message:
NAs introduced by coercion

> num1 <- as.integer(readline(prompt = "enter the first number:"))
enter the first number:20
> num2 <- as.integer(readline(prompt = "enter the second number:"))
enter the second number:10
> num3 <- num1 + num2
> print(num3)
[1] 30
>
```

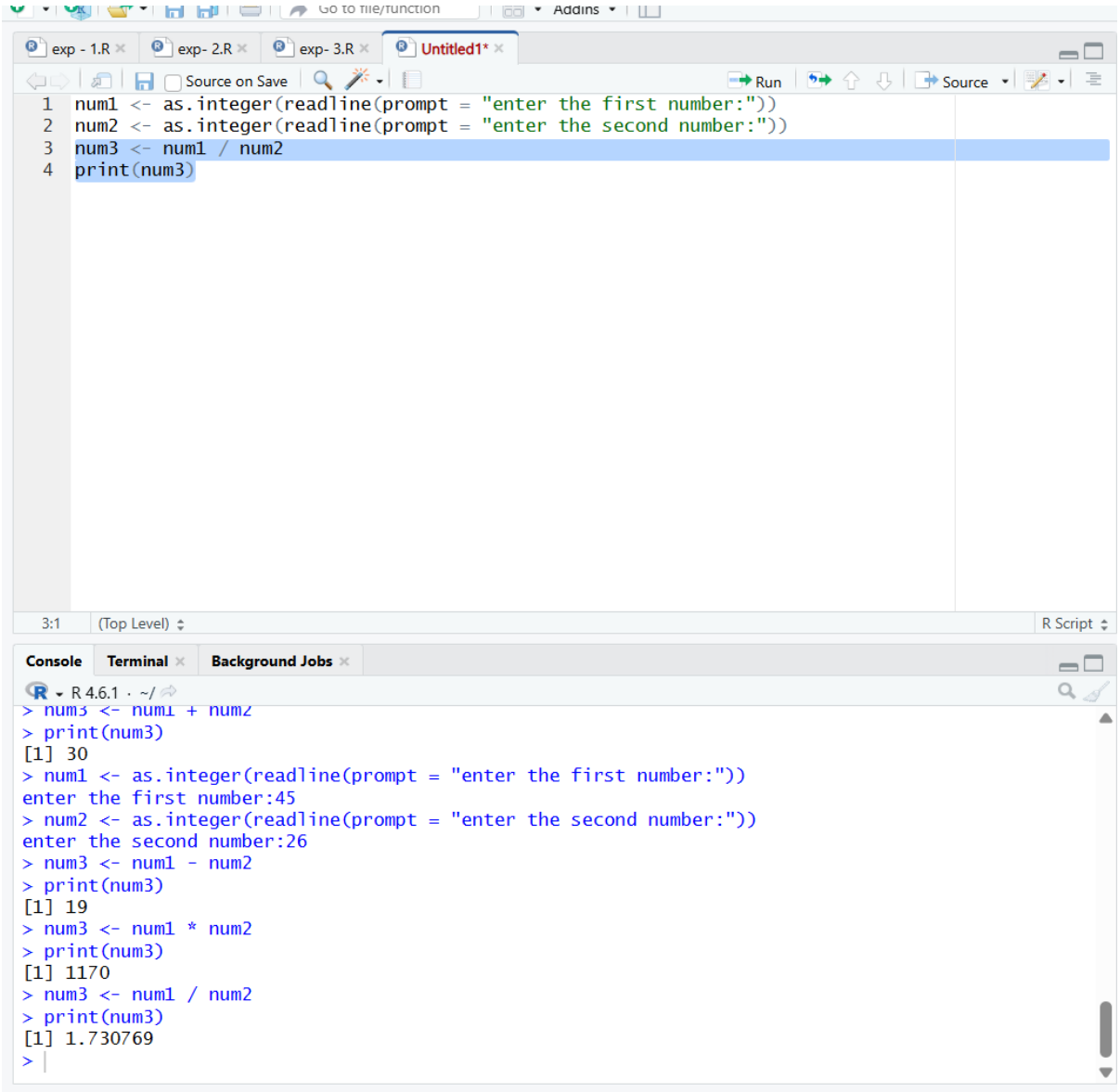
Exp - 2


exp - 1.R
Untitled1*
Source on Save
Run
Source
1 num1 <- as.integer(readline(prompt = "enter the first number:"))
2 num2 <- as.integer(readline(prompt = "enter the second number:"))
3 num3 <- num1 - num2
4 print(num3)
3:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> num1 <- as.integer(readline(prompt = "enter the first number:"))
enter the first number:20
> num2 <- as.integer(readline(prompt = "enter the second number:"))
enter the second number:10
> num3 <- num1 + num2
> print(num3)
[1] 30
> num1 <- as.integer(readline(prompt = "enter the first number:"))
enter the first number:45
> num2 <- as.integer(readline(prompt = "enter the second number:"))
enter the second number:26
> num3 <- num1 - num2
> print(num3)
[1] 19

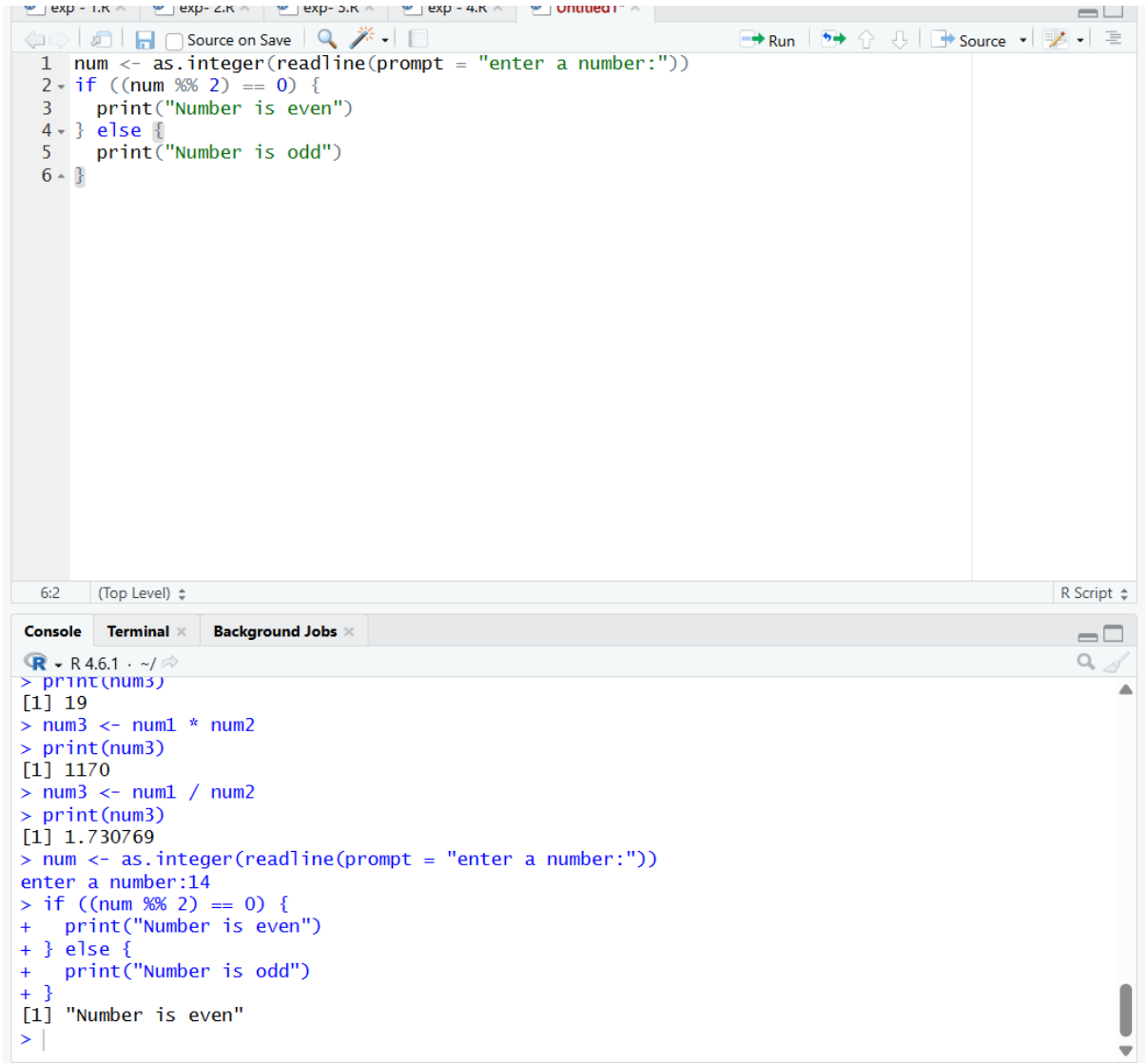
Exp- 3


Source on Save
Run
Source
1 num1 <- as.integer(readline(prompt = "enter the first number:"))
2 num2 <- as.integer(readline(prompt = "enter the second number:"))
3 num3 <- num1 * num2
4 print(num3)
3:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
enter the first number:20
> num2 <- as.integer(readline(prompt = "enter the second number:"))
enter the second number:10
> num3 <- num1 + num2
> print(num3)
[1] 30
> num1 <- as.integer(readline(prompt = "enter the first number:"))
enter the first number:45
> num2 <- as.integer(readline(prompt = "enter the second number:"))
enter the second number:26
> num3 <- num1 - num2
> print(num3)
[1] 19
> num3 <- num1 * num2
> print(num3)
[1] 1170
>

Exp - 4


exp - 1.R
exp- 2.R
exp- 3.R
Untitled1*
Source on Save
Run
Source
num1 <- as.integer(readline(prompt = "enter the first number:"))
num2 <- as.integer(readline(prompt = "enter the second number:"))
num3 <- num1 / num2
print(num3)
3:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> num3 <- num1 + num2
> print(num3)
[1] 30
> num1 <- as.integer(readline(prompt = "enter the first number:"))
enter the first number:45
> num2 <- as.integer(readline(prompt = "enter the second number:"))
enter the second number:26
> num3 <- num1 - num2
> print(num3)
[1] 19
> num3 <- num1 * num2
> print(num3)
[1] 1170
> num3 <- num1 / num2
> print(num3)
[1] 1.730769
>

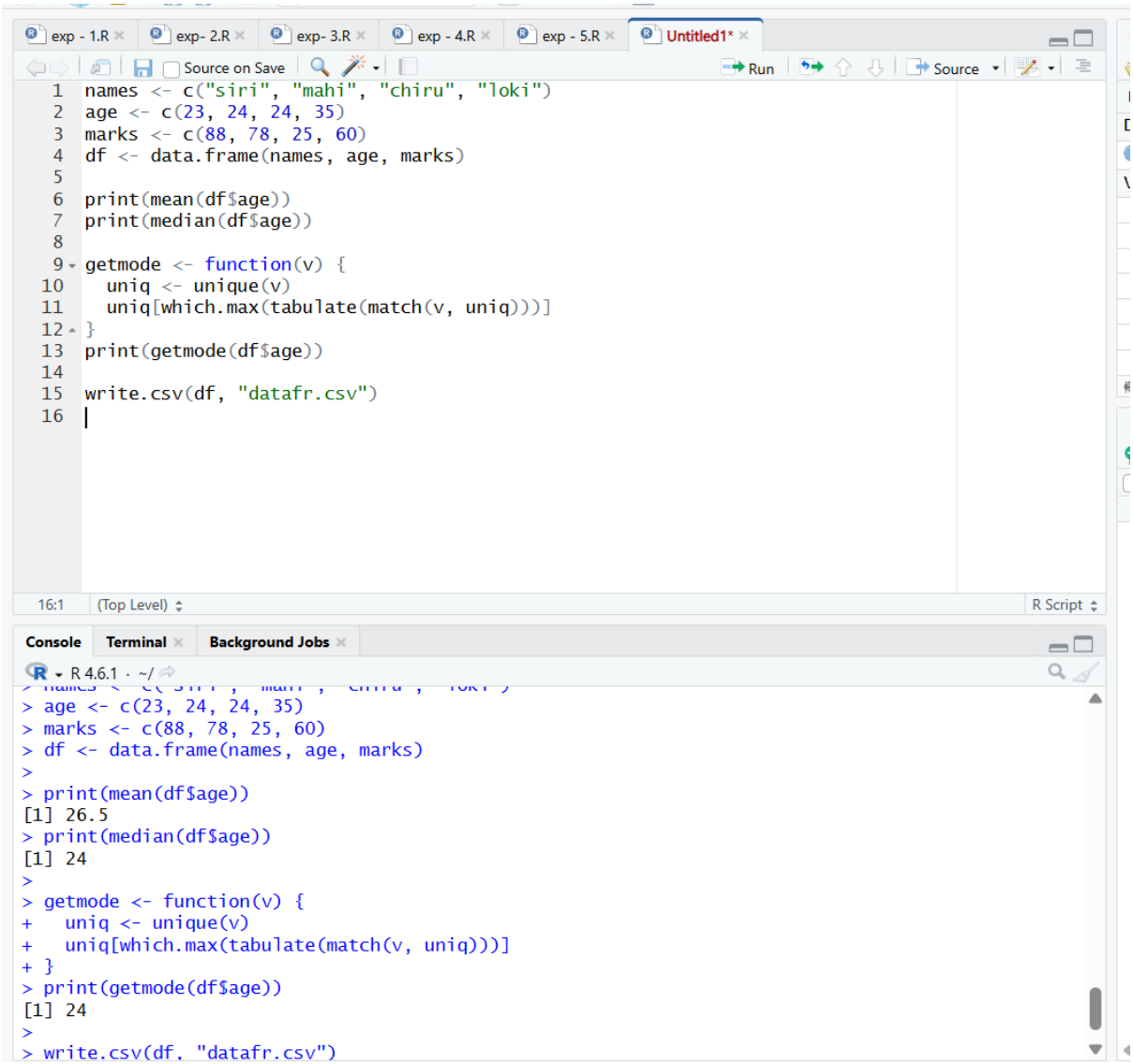
Exp – 5



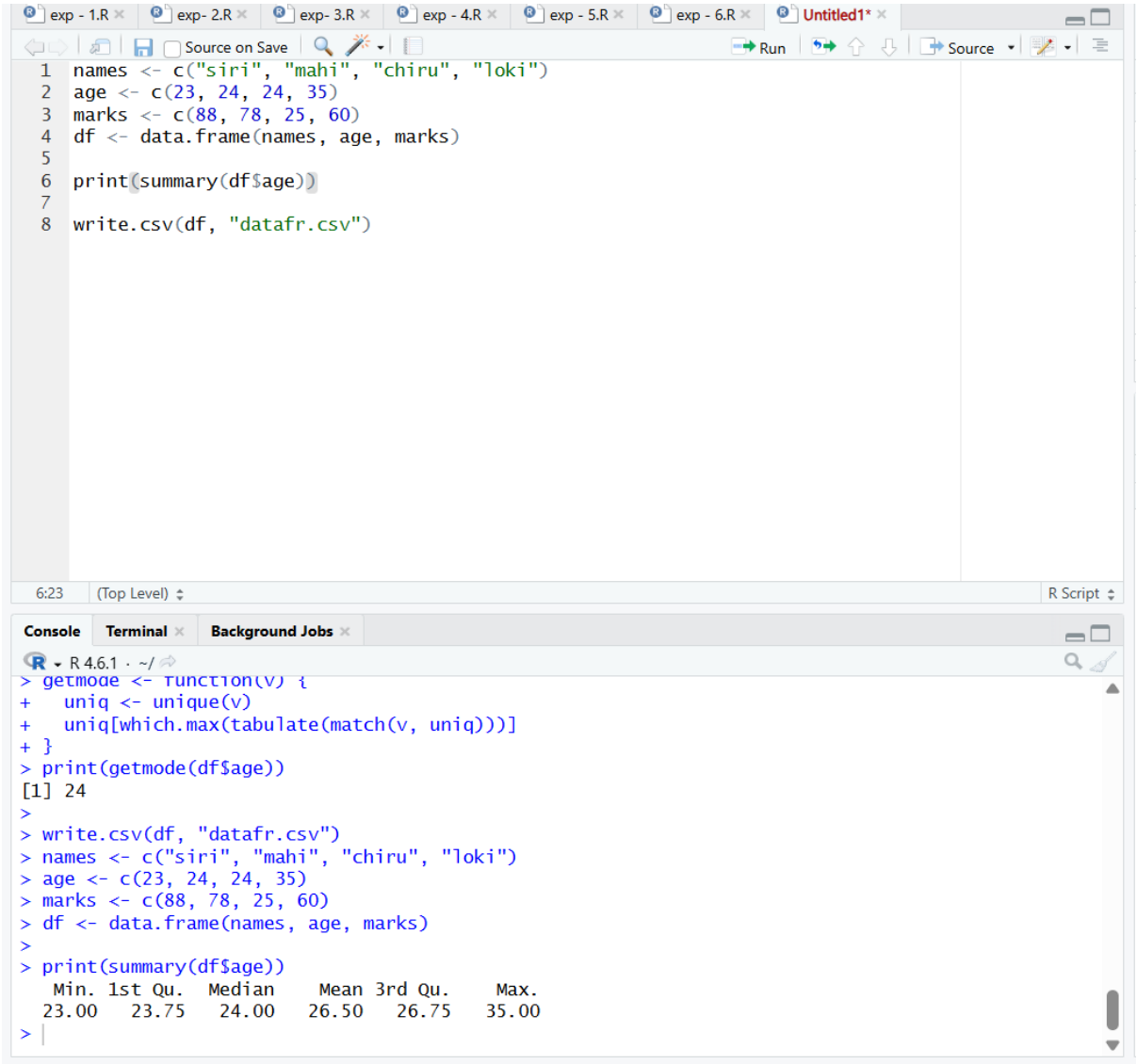
```
num <- as.integer(readline(prompt = "enter a number:"))
if ((num %% 2) == 0) {
  print("Number is even")
} else {
  print("Number is odd")
}
```

```
> print(num3)
[1] 19
> num3 <- num1 * num2
> print(num3)
[1] 1170
> num3 <- num1 / num2
> print(num3)
[1] 1.730769
> num <- as.integer(readline(prompt = "enter a number:"))
enter a number:14
> if ((num %% 2) == 0) {
+   print("Number is even")
+ } else {
+   print("Number is odd")
+ }
[1] "Number is even"
>
```

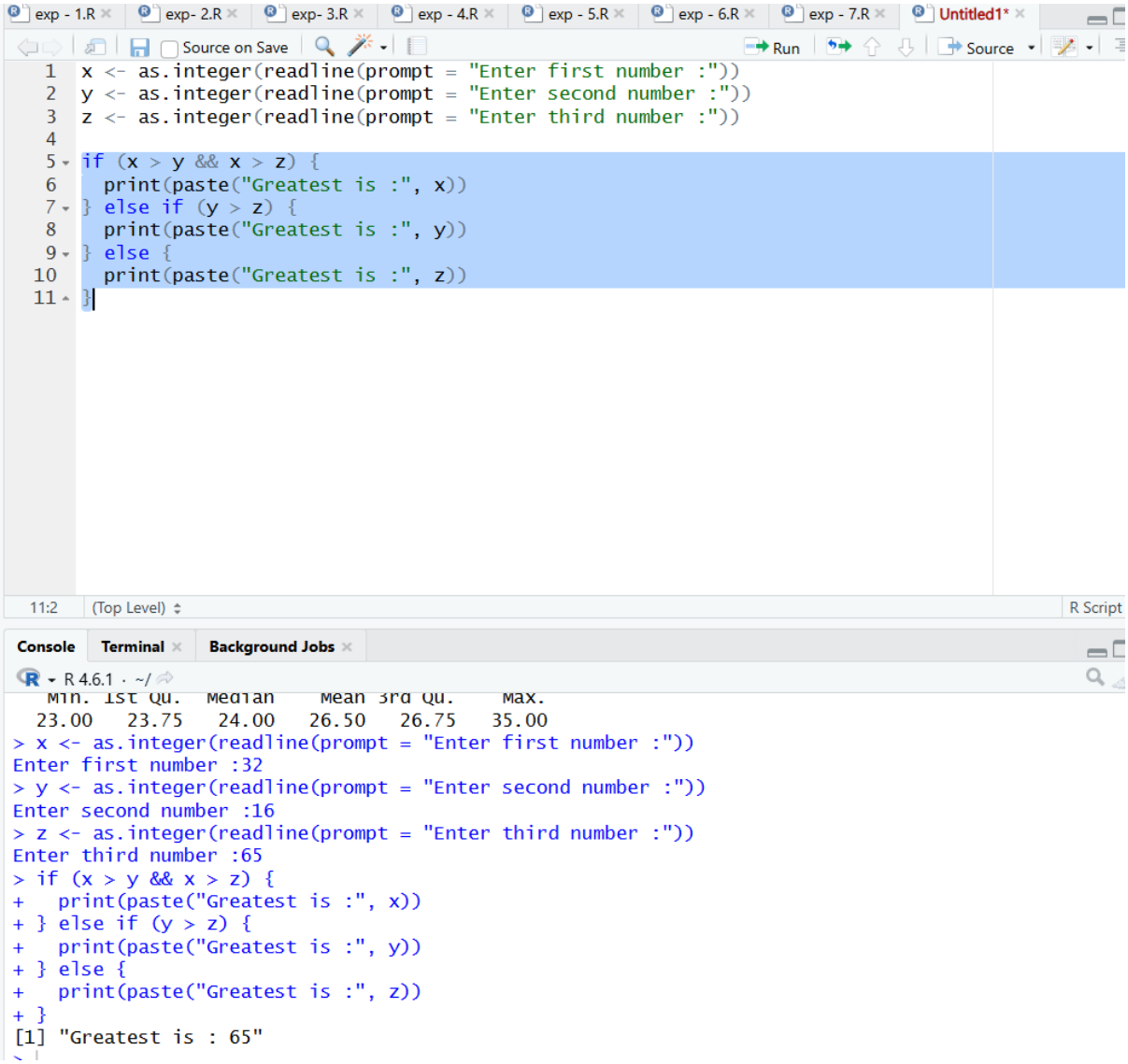
Exp – 6


exp - 1.R
exp- 2.R
exp- 3.R
exp - 4.R
exp - 5.R
Untitled1*
Source on Save
Run
Source
names <- c("siri", "mahi", "chiru", "loki")
age <- c(23, 24, 24, 35)
marks <- c(88, 78, 25, 60)
df <- data.frame(names, age, marks)
print(mean(df$age))
print(median(df$age))
getmode <- function(v) {
uniq <- unique(v)
uniq[which.max(tabulate(match(v, uniq)))]
}
print(getmode(df$age))
write.csv(df, "datafr.csv")
16:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> age <- c(23, 24, 24, 35)
> marks <- c(88, 78, 25, 60)
> df <- data.frame(names, age, marks)
>
> print(mean(df$age))
[1] 26.5
> print(median(df$age))
[1] 24
>
> getmode <- function(v) {
+ uniq <- unique(v)
+ uniq[which.max(tabulate(match(v, uniq)))]
+ }
> print(getmode(df$age))
[1] 24
>
> write.csv(df, "datafr.csv")

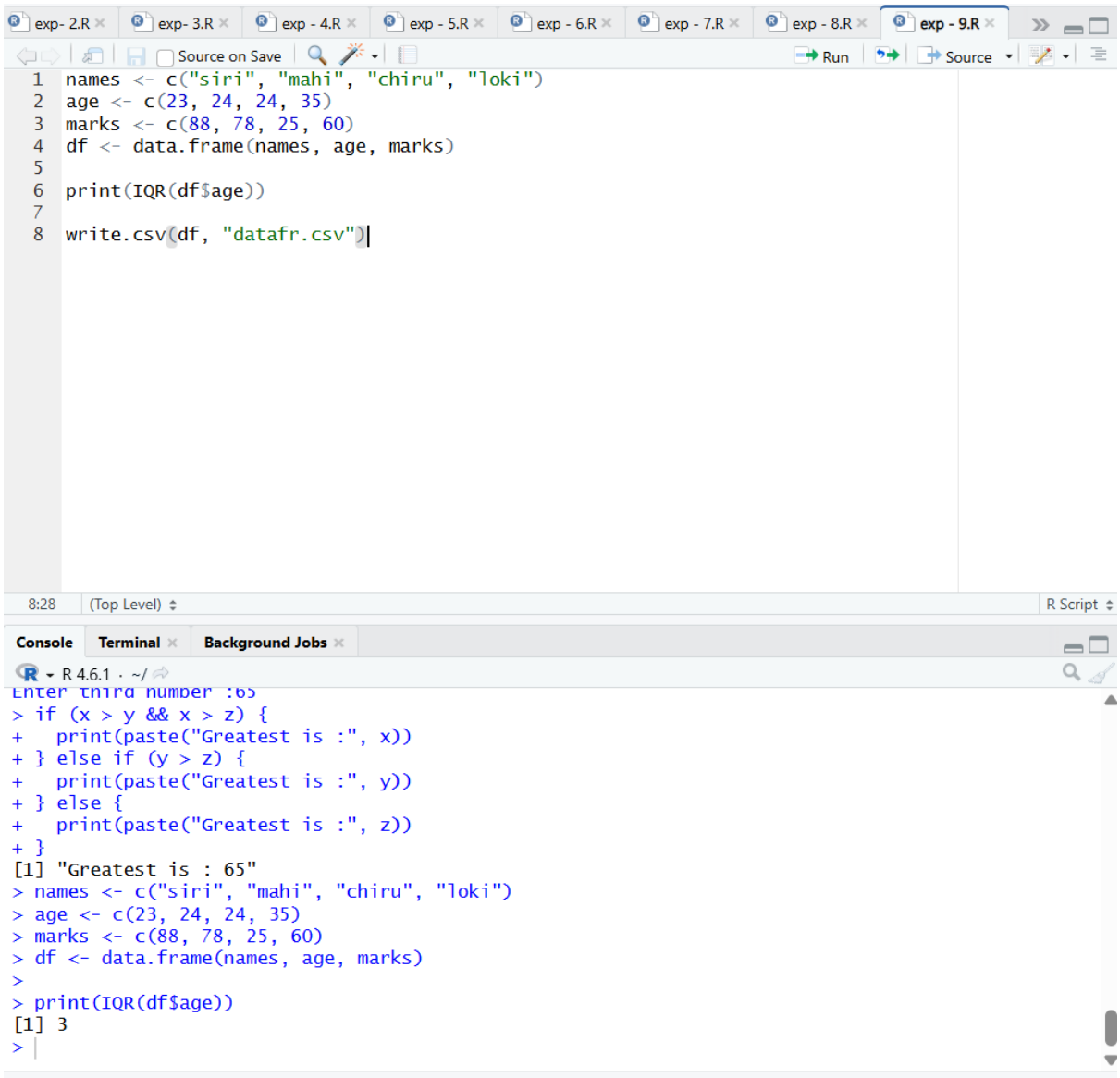
Exp 7


exp - 1.R
exp- 2.R
exp- 3.R
exp - 4.R
exp - 5.R
exp - 6.R
Untitled1*
Source on Save
Run
Source
1 names <- c("siri", "mahi", "chiru", "loki")
2 age <- c(23, 24, 24, 35)
3 marks <- c(88, 78, 25, 60)
4 df <- data.frame(names, age, marks)
5
6 print(summary(df$age))
7
8 write.csv(df, "datafr.csv")
6:23
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> getmode <- function(v) {
+ uniq <- unique(v)
+ uniq[which.max(tabulate(match(v, uniq)))]
+ }
> print(getmode(df$age))
[1] 24
>
> write.csv(df, "datafr.csv")
> names <- c("siri", "mahi", "chiru", "loki")
> age <- c(23, 24, 24, 35)
> marks <- c(88, 78, 25, 60)
> df <- data.frame(names, age, marks)
>
> print(summary(df$age))
Min. 1st Qu. Median Mean 3rd Qu. Max.
23.00 23.75 24.00 26.50 26.75 35.00
>

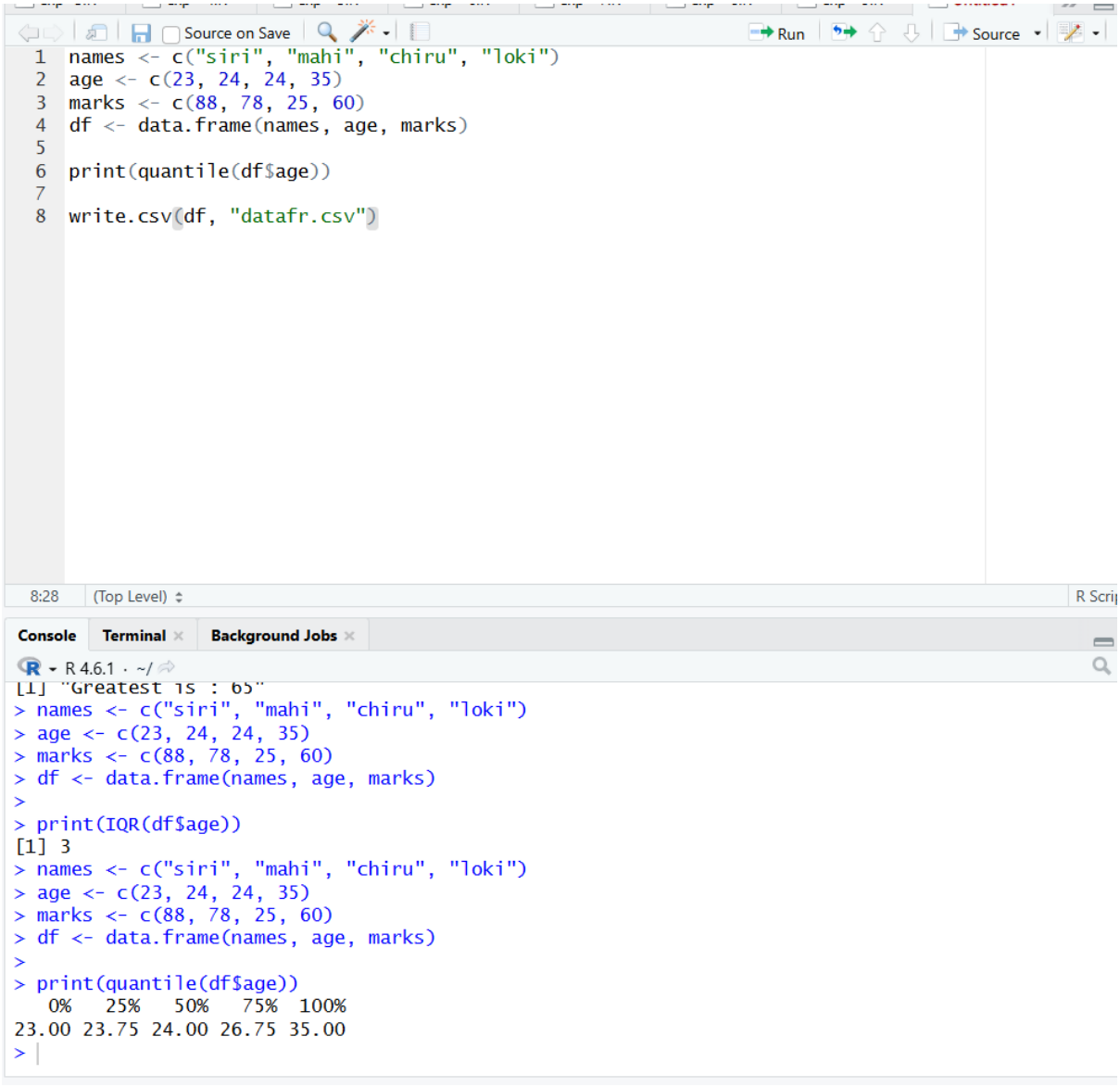
Exp 8


exp - 1.R
exp- 2.R
exp- 3.R
exp - 4.R
exp - 5.R
exp - 6.R
exp - 7.R
Untitled1*
Source on Save
Run
Source
1 x <- as.integer(readline(prompt = "Enter first number :"))
2 y <- as.integer(readline(prompt = "Enter second number :"))
3 z <- as.integer(readline(prompt = "Enter third number :"))
4
5 if (x > y && x > z) {
6 print(paste("Greatest is :", x))
7 } else if (y > z) {
8 print(paste("Greatest is :", y))
9 } else {
10 print(paste("Greatest is :", z))
11 }
11:2
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
Min. 1st Qu. Median Mean 3rd Qu. Max.
23.00 23.75 24.00 26.50 26.75 35.00
> x <- as.integer(readline(prompt = "Enter first number :"))
Enter first number :32
> y <- as.integer(readline(prompt = "Enter second number :"))
Enter second number :16
> z <- as.integer(readline(prompt = "Enter third number :"))
Enter third number :65
> if (x > y && x > z) {
+ print(paste("Greatest is :", x))
+ } else if (y > z) {
+ print(paste("Greatest is :", y))
+ } else {
+ print(paste("Greatest is :", z))
+ }
[1] "Greatest is : 65"

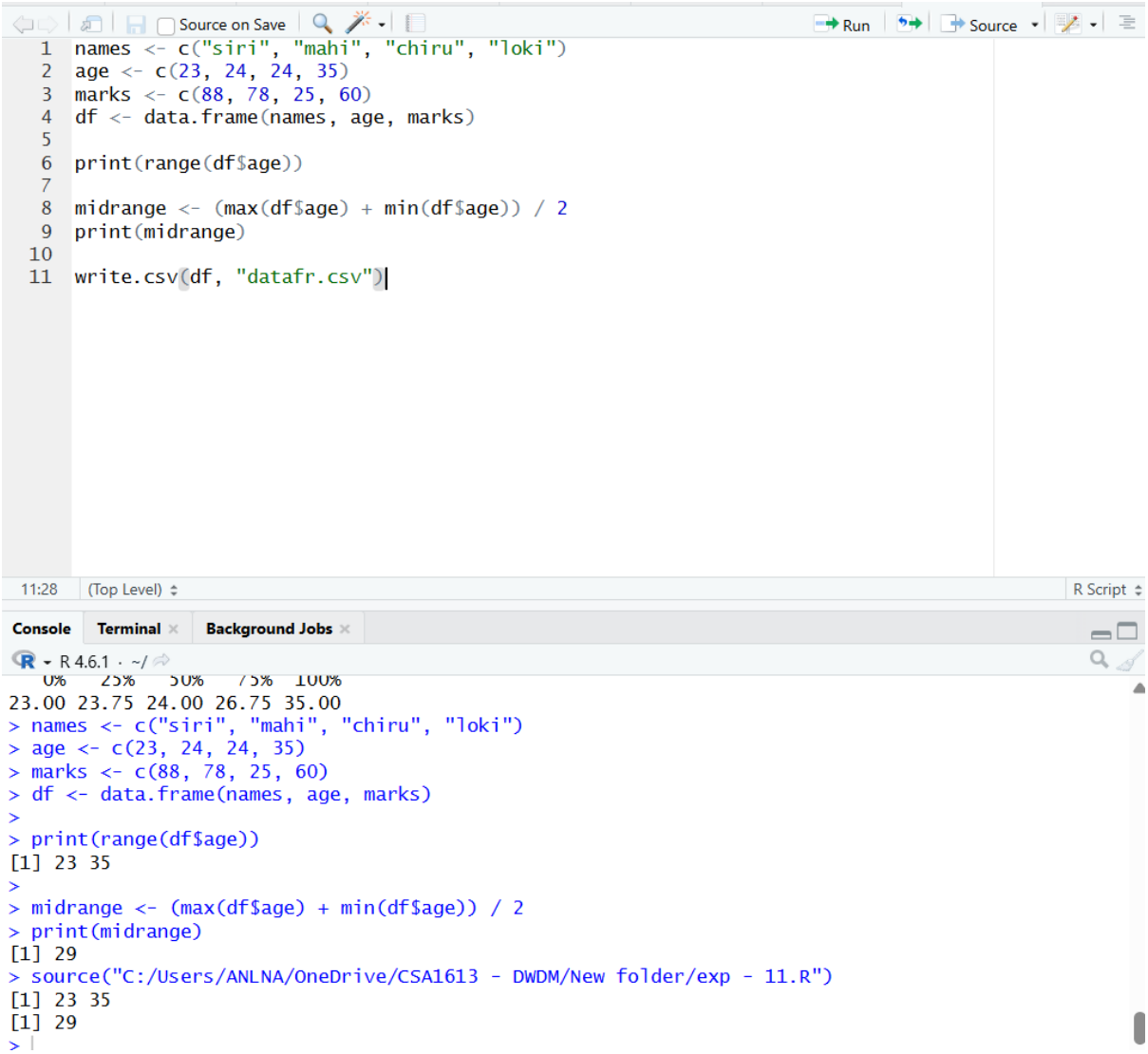
Exp 9


exp- 2.R
exp- 3.R
exp - 4.R
exp - 5.R
exp - 6.R
exp - 7.R
exp - 8.R
exp - 9.R
Source on Save
Run
Source
names <- c("siri", "mahi", "chiru", "loki")
age <- c(23, 24, 24, 35)
marks <- c(88, 78, 25, 60)
df <- data.frame(names, age, marks)
print(IQR(df$age))
write.csv(df, "datafr.csv")
8:28
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
Enter third number :65
> if (x > y && x > z) {
+ print(paste("Greatest is :", x))
+ } else if (y > z) {
+ print(paste("Greatest is :", y))
+ } else {
+ print(paste("Greatest is :", z))
+ }
[1] "Greatest is : 65"
> names <- c("siri", "mahi", "chiru", "loki")
> age <- c(23, 24, 24, 35)
> marks <- c(88, 78, 25, 60)
> df <- data.frame(names, age, marks)
>
> print(IQR(df$age))
[1] 3
>

Exp 10


Source on Save
Run
Source
1 names <- c("siri", "mahi", "chiru", "loki")
2 age <- c(23, 24, 24, 35)
3 marks <- c(88, 78, 25, 60)
4 df <- data.frame(names, age, marks)
5
6 print(quantile(df$age))
7
8 write.csv(df, "datafr.csv")
8:28
(Top Level)
Console
Terminal
Background Jobs
R 4.6.1 · ~/
[1] "Greatest is : 65"
> names <- c("siri", "mahi", "chiru", "loki")
> age <- c(23, 24, 24, 35)
> marks <- c(88, 78, 25, 60)
> df <- data.frame(names, age, marks)
>
> print(IQR(df$age))
[1] 3
> names <- c("siri", "mahi", "chiru", "loki")
> age <- c(23, 24, 24, 35)
> marks <- c(88, 78, 25, 60)
> df <- data.frame(names, age, marks)
>
> print(quantile(df$age))
0% 25% 50% 75% 100%
23.00 23.75 24.00 26.75 35.00
>

Exp 11



```
names <- c("siri", "mahi", "chiru", "loki")
age <- c(23, 24, 24, 35)
marks <- c(88, 78, 25, 60)
df <- data.frame(names, age, marks)

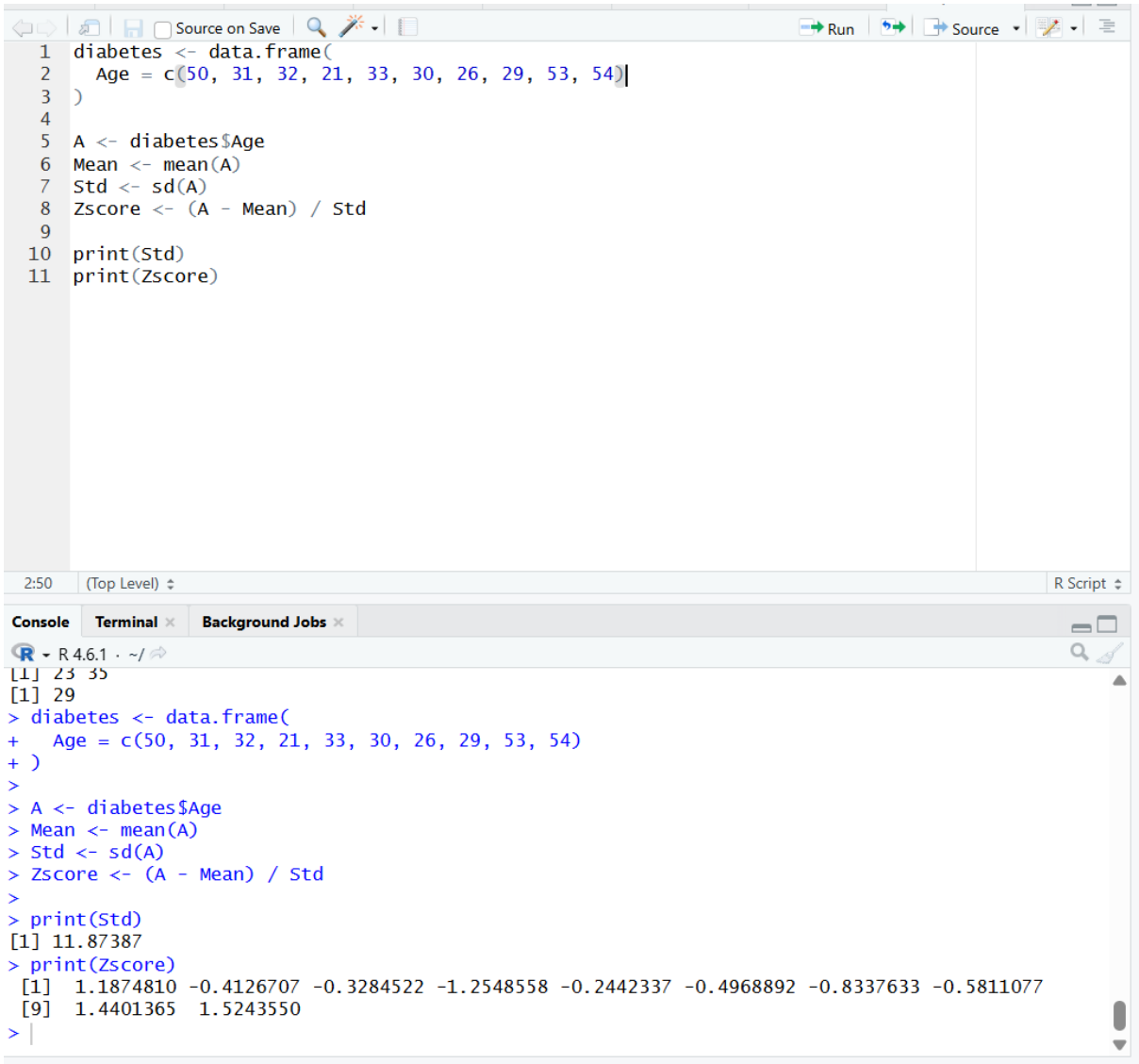
print(range(df$age))

midrange <- (max(df$age) + min(df$age)) / 2
print(midrange)

write.csv(df, "datafr.csv")
```

```
   0%   25%   50%   75%  100%
23.00 23.75 24.00 26.75 35.00
> names <- c("siri", "mahi", "chiru", "loki")
> age <- c(23, 24, 24, 35)
> marks <- c(88, 78, 25, 60)
> df <- data.frame(names, age, marks)
>
> print(range(df$age))
[1] 23 35
>
> midrange <- (max(df$age) + min(df$age)) / 2
> print(midrange)
[1] 29
> source("C:/Users/ANLNA/OneDrive/CSA1613 - DWDM/New folder/exp - 11.R")
[1] 23 35
[1] 29
>
```

Exp 12


Source on Save
Run
Source
1 diabetes <- data.frame(
2 Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54)
3)
4
5 A <- diabetes$Age
6 Mean <- mean(A)
7 Std <- sd(A)
8 Zscore <- (A - Mean) / Std
9
10 print(Std)
11 print(Zscore)
2:50
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
[1] 23 35
[1] 29
> diabetes <- data.frame(
+ Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54)
+)
>
> A <- diabetes$Age
> Mean <- mean(A)
> Std <- sd(A)
> Zscore <- (A - Mean) / Std
>
> print(Std)
[1] 11.87387
> print(Zscore)
[1] 1.1874810 -0.4126707 -0.3284522 -1.2548558 -0.2442337 -0.4968892 -0.8337633 -0.5811077
[9] 1.4401365 1.5243550
>

Exp 13

```r
diabetes <- data.frame(
  Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54)
)

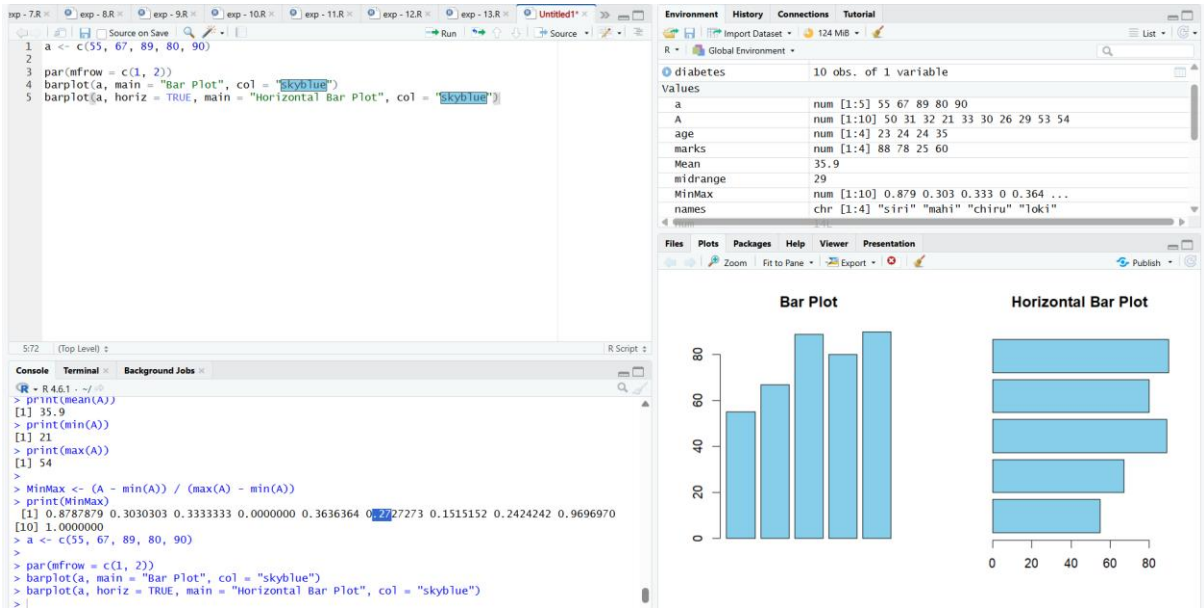
A <- diabetes$Age

print(mean(A))
print(min(A))
print(max(A))

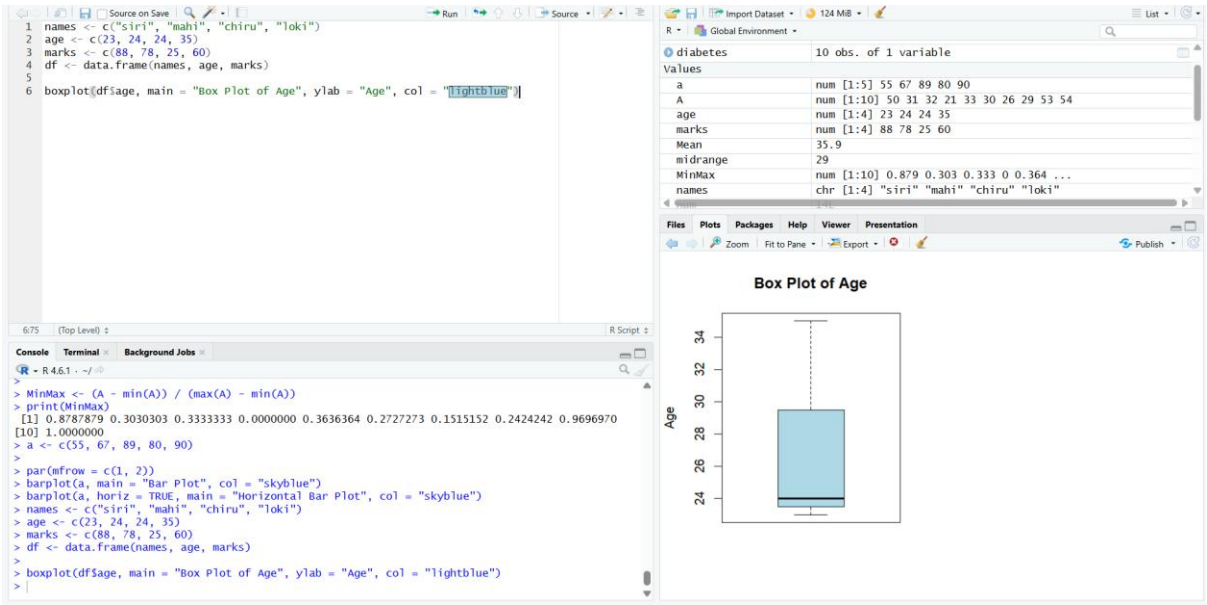
MinMax <- (A - min(A)) / (max(A) - min(A))
print(MinMax)
```

```
+   Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54)
+ )
>
> A <- diabetes$Age
>
> print(mean(A))
[1] 35.9
> print(min(A))
[1] 21
> print(max(A))
[1] 54
>
> MinMax <- (A - min(A)) / (max(A) - min(A))
> print(MinMax)
 [1] 0.8787879 0.3030303 0.3333333 0.0000000 0.3636364 0.2727273 0.1515152 0.2424242 0.9696970
[10] 1.0000000
>
```

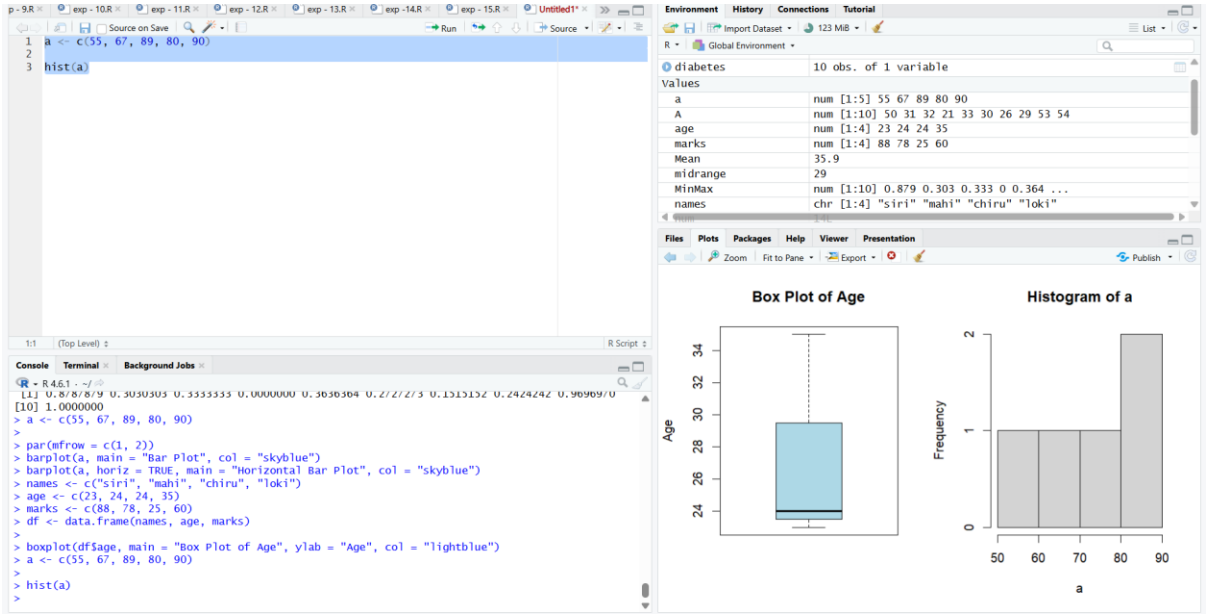
Exp 14



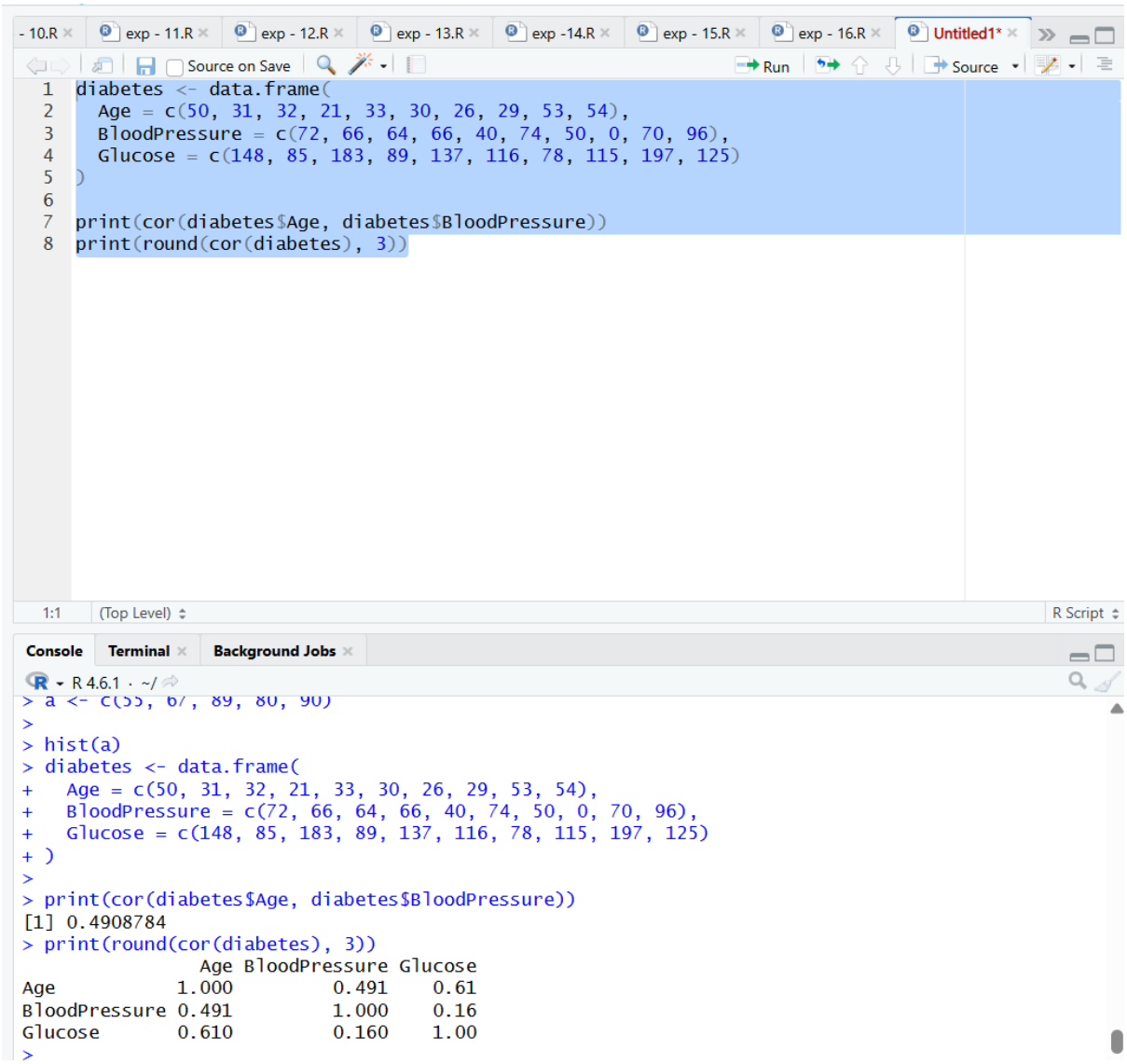
Exp 15


names <- c("siri", "mahi", "chiru", "loki")
age <- c(23, 24, 24, 35)
marks <- c(88, 78, 25, 60)
df <- data.frame(names, age, marks)
boxplot(df$age, main = "Box Plot of Age", ylab = "Age", col = "lightblue")
> MinMax <- (A - min(A)) / (max(A) - min(A))
> print(MinMax)
[1] 0.8787879 0.3030303 0.3333333 0.0000000 0.3636364 0.2727273 0.1515152 0.2424242 0.9696970
[10] 1.0000000
> a <- c(55, 67, 89, 80, 90)
> par(mfrow = c(1, 2))
> barplot(a, main = "Bar Plot", col = "skyblue")
> barplot(a, horiz = TRUE, main = "Horizontal Bar Plot", col = "skyblue")
> names <- c("siri", "mahi", "chiru", "loki")
> age <- c(23, 24, 24, 35)
> marks <- c(88, 78, 25, 60)
> df <- data.frame(names, age, marks)
> boxplot(df$age, main = "Box Plot of Age", ylab = "Age", col = "lightblue")
diabetes 10 obs. of 1 variable
Values
a num [1:5] 55 67 89 80 90
A num [1:10] 50 31 32 21 33 30 26 29 53 54
age num [1:4] 23 24 24 35
marks num [1:4] 88 78 25 60
Mean 35.9
midrange 29
MinMax num [1:10] 0.879 0.303 0.333 0 0.364 ...
names chr [1:4] "siri" "mahi" "chiru" "loki"
Box Plot of Age
Age
24
26
28
30
32
34

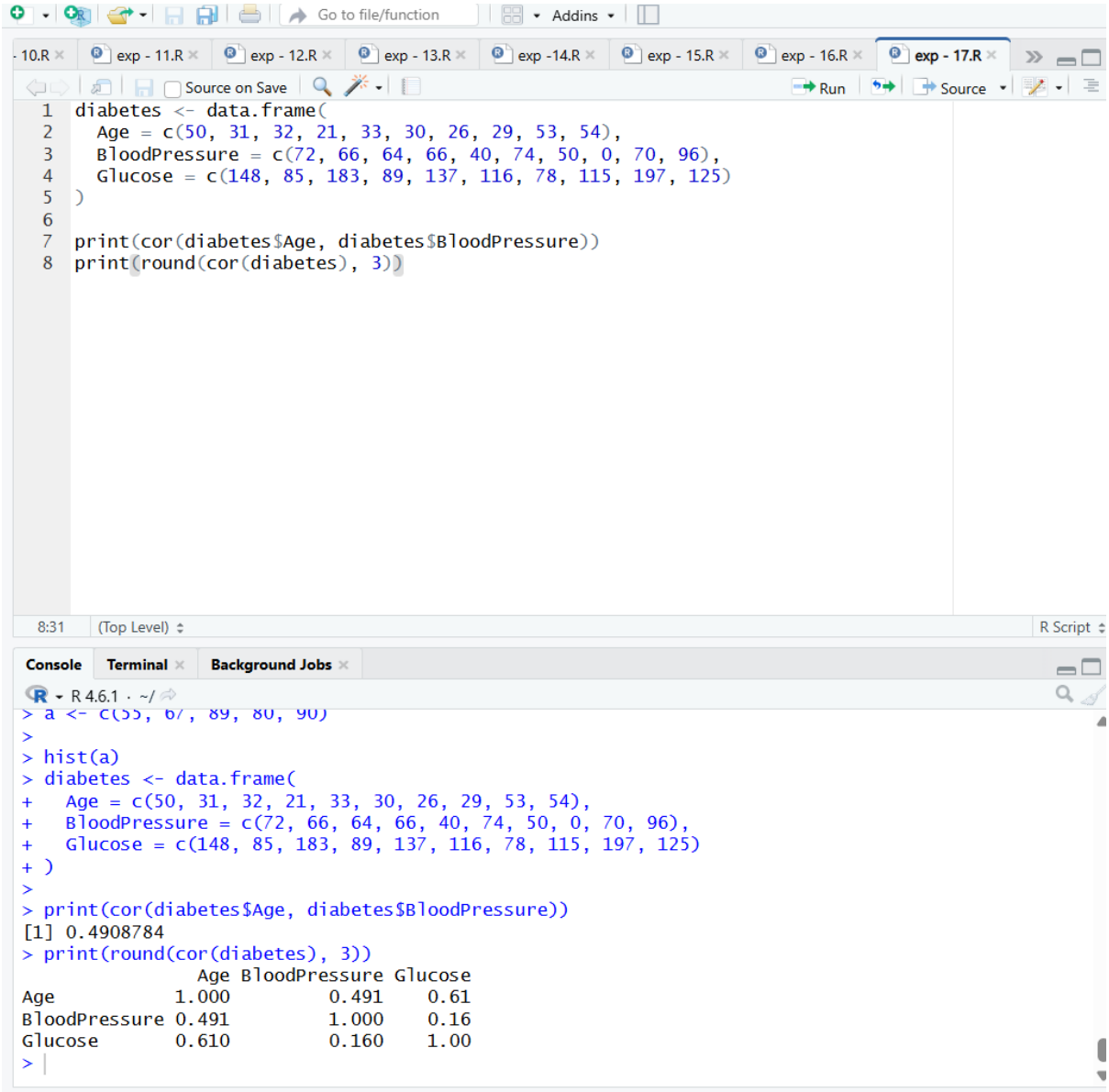
Exp 16


a <- c(55, 67, 89, 80, 90)
hist(a)
> a <- c(55, 67, 89, 80, 90)
> par(mfrow = c(1, 2))
> barplot(a, main = "Bar Plot", col = "skyblue")
> barplot(a, horiz = TRUE, main = "Horizontal Bar Plot", col = "skyblue")
> names <- c("siri", "mahi", "chiru", "loki")
> age <- c(23, 24, 24, 35)
> marks <- c(88, 78, 25, 60)
> df <- data.frame(names, age, marks)
> boxplot(df$age, main = "Box Plot of Age", ylab = "Age", col = "lightblue")
> a <- c(55, 67, 89, 80, 90)
> hist(a)
Box Plot of Age
Histogram of a
Frequency
0
1
2
50
60
70
80
90
a

Exp 17


- 10.R
exp - 11.R
exp - 12.R
exp - 13.R
exp -14.R
exp - 15.R
exp - 16.R
Untitled1*
Source on Save
Run
Source
1 diabetes <- data.frame(
2 Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54),
3 BloodPressure = c(72, 66, 64, 66, 40, 74, 50, 0, 70, 96),
4 Glucose = c(148, 85, 183, 89, 137, 116, 78, 115, 197, 125)
5)
6
7 print(cor(diabetes$Age, diabetes$BloodPressure))
8 print(round(cor(diabetes), 3))
1:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> a <- c(55, 67, 89, 80, 90)
>
> hist(a)
> diabetes <- data.frame(
+ Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54),
+ BloodPressure = c(72, 66, 64, 66, 40, 74, 50, 0, 70, 96),
+ Glucose = c(148, 85, 183, 89, 137, 116, 78, 115, 197, 125)
+)
>
> print(cor(diabetes$Age, diabetes$BloodPressure))
[1] 0.4908784
> print(round(cor(diabetes), 3))
Age BloodPressure Glucose
Age 1.000 0.491 0.61
BloodPressure 0.491 1.000 0.16
Glucose 0.610 0.160 1.00
>

Exp 18


Go to file/function
Addins
10.R
exp - 11.R
exp - 12.R
exp - 13.R
exp -14.R
exp - 15.R
exp - 16.R
exp - 17.R
Source on Save
Run
Source
1 diabetes <- data.frame(
2 Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54),
3 BloodPressure = c(72, 66, 64, 66, 40, 74, 50, 0, 70, 96),
4 Glucose = c(148, 85, 183, 89, 137, 116, 78, 115, 197, 125)
5)
6
7 print(cor(diabetes$Age, diabetes$BloodPressure))
8 print(round(cor(diabetes), 3))
8:31
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> a <- c(55, 67, 89, 80, 90)
>
> hist(a)
> diabetes <- data.frame(
+ Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54),
+ BloodPressure = c(72, 66, 64, 66, 40, 74, 50, 0, 70, 96),
+ Glucose = c(148, 85, 183, 89, 137, 116, 78, 115, 197, 125)
+)
>
> print(cor(diabetes$Age, diabetes$BloodPressure))
[1] 0.4908784
> print(round(cor(diabetes), 3))
Age BloodPressure Glucose
Age 1.000 0.491 0.61
BloodPressure 0.491 1.000 0.16
Glucose 0.610 0.160 1.00
>

Exp 19

```
diabetes <- data.frame(
  Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54),
  BloodPressure = c(72, 66, 64, 66, 40, 74, 50, 0, 70, 96),
  Glucose = c(148, 85, 183, 89, 137, 116, 78, 115, 197, 125)
)

plot(diabetes$Age, diabetes$Glucose,
     main = "Scatter Plot of Age vs Glucose",
     xlab = "Age", ylab = "Glucose",
     col = "blue", pch = 19)
```

```
> print(round(cor(diabetes), 3))
              Age BloodPressure Glucose
Age           1.000         0.491    0.61
BloodPressure 0.491         1.000    0.16
Glucose       0.610         0.160    1.00
> diabetes <- data.frame(
+   Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54),
+   BloodPressure = c(72, 66, 64, 66, 40, 74, 50, 0, 70, 96),
+   Glucose = c(148, 85, 183, 89, 137, 116, 78, 115, 197, 125)
+ )
>
> plot(diabetes$Age, diabetes$Glucose,
+      main = "Scatter Plot of Age vs Glucose",
+      xlab = "Age", ylab = "Glucose",
+      col = "blue", pch = 19)
>
```

Environment	
diabetes	10 obs. of 3 variables
Values	
a	num [1:5] 55 67 89 80 90
A	num [1:10] 50 31 32 21 33 30 26 29 53 54
age	num [1:4] 23 24 24 35
marks	num [1:4] 88 78 25 60
Mean	35.9
midrange	29
MinMax	num [1:10] 0.879 0.303 0.333 0 0.364 ...
names	chr [1:4] "siri" "mahi" "chiru" "loki"



Exp 20

```
diabetes <- data.frame(
  Age = c(50, 31, 32, 21, 33, 30, 26, 29, 53, 54),
  BloodPressure = c(72, 66, 64, 66, 40, 74, 50, 0, 70, 96),
  Glucose = c(148, 85, 183, 89, 137, 116, 78, 115, 197, 125)
)

Relation <- lm(BloodPressure ~ Age, data = diabetes)
print(Relation)

plot(diabetes$Age, diabetes$BloodPressure,
     col = "darkgreen",
     main = "Linear Regression Analysis",
     xlab = "Age", ylab = "BloodPressure")
abline(Relation, col = "red")
```

```
> Relation <- lm(BloodPressure ~ Age, data = diabetes)
> print(Relation)

Call:
lm(formula = BloodPressure ~ Age, data = diabetes)

Coefficients:
(Intercept)          Age
     21.724        1.061

>
> plot(diabetes$Age, diabetes$BloodPressure,
+      col = "darkgreen",
+      main = "Linear Regression Analysis",
+      xlab = "Age", ylab = "BloodPressure")
> abline(Relation, col = "red")
>
```

Environment	
diabetes	10 obs. of 3 variables
Relation	List of 12
Values	
a	num [1:5] 55 67 89 80 90
A	num [1:10] 50 31 32 21 33 30 26 29 53 54
age	num [1:4] 23 24 24 35
marks	num [1:4] 88 78 25 60
Mean	35.9
midrange	29
MinMax	num [1:10] 0.879 0.303 0.333 0 0.364 ...

